ID	Tier	Category	Metric	Measure / threshold	Calibrate ?
T1	C	Text difficulty	Flesch-Kincaid grade level	Target ~3-6 on prose blocks only (exclude nav, lists, tables, headings); report with T2 as a range; flag if the two differ by > 2 grades	No
T2	C	Text difficulty	Dale-Chall (or Spache for Grade 3)	Dale-Chall <= 5.9 (grade 5-6), <= 4.9 (grade <= 4); Spache ~3-4	No
T3	C	Text difficulty	Sentence length	Flag > 25 words; ideal 15-20 words	No
T4	C	Text difficulty	Conjunctions per sentence	<= 2	No
T5	D	Text difficulty	Nested / subordinate clauses	Flag sentences with >= 2 subordinate clauses (dependency parser)	Yes
T6	C	Text difficulty	Passive voice	Flag; no sourced number	n/a (report only)
T7	C	Text difficulty	Rare / unfamiliar words	% of tokens outside a common-word list (COGA: “the most common 1500 words or phrases”; Dale–Chall’s 3,000 familiar words is a published alternative), proper nouns excluded; flag > 10 % (cal.); uncommon terms need an explanation within one click	Yes
T8	D	Text difficulty	Unexplained abbreviations	All-caps token with no <abbr>, expansion at first use, or glossary entry -> 0 allowed	No
T9	D	Text difficulty	Jargon / curriculum terms without definition	Term outside common-word list and not defined on page or within one click	Yes
T10	D	Text difficulty	Reading level (overall)	WCAG 3.1.5 AAA = lower secondary; project target is stricter (Grade 3-5)	n/a (report only)
T11	D	Text difficulty	Lexile / ATOS / TextEvaluator (optional, external)	Lexile G3 520-820L, G4 740-940L, G5 830-1010L; TextEvaluator best validated (rho .756)	No
S1	C	Structure and chunking	Paragraph length	Flag > 50 words (COGA) or > 5 sentences (Ontario.ca); Canada.ca: <= 3 sentences	No
S2	C	Structure and chunking	Heading density	>= 1 heading per ~200 words	No

ID	Tier	Category	Metric	Measure / threshold	Calibrate ?
S3	C	Structure and chunking	Section headings present	Every content section has a heading	No
S4	D	Structure and chunking	List used for 3+ items or steps	Flag sentences enumerating >= 3 items inline	Yes
S5	C	Structure and chunking	List length	<= 7 items; one sentence per item	No
S6	D	Structure and chunking	Steps / instructions	One action per step; numbered when order matters; children skip long instructions	Yes
S7	D	Structure and chunking	Front-loaded headings	Key noun within the first two words	No
J1	D	Judgment check	Headings too similar	Pairwise sentence-embedding cosine > 0.8, or >= 60 % shared lemmas	Yes
J2	D	Judgment check	Heading generic / non-descriptive	Stoplist ('Introduction', 'More', 'Part 2') or similarity to own section body < 0.3	Yes
J3	D	Judgment check	Heading does not match section content	Similarity heading <-> section text < 0.3	Yes
J4	D	Judgment check	Page title vs h1 mismatch	Similarity < 0.5	Yes
J5	C	Judgment check	Generic link text	'click here', 'more', 'here', 'read more' -> 0 allowed	No
J6	E	Judgment check	Link text vs destination	Similarity link text <-> target page title/h1 < 0.3	Yes
J7	E	Judgment check	Same label, different targets	One link text -> more than one href	No
J8	E	Judgment check	Different labels, same target	One href -> more than one label across pages	No
J9	D	Judgment check	Navigation labels too similar to each other	Pairwise cosine > 0.8	Yes

ID	Tier	Category	Metric	Measure / threshold	Calibrate ?
J10	D	Judgment check	Icon unrelated to label / function	Image-text similarity (CLIP) below reference median, or LLM-judge disagreement with label	Yes
J11	C	Judgment check	Icon without text label	Control whose only content is an image/SVG -> flag all	No
J12	D	Judgment check	Alt text quality	Alt equals filename, 'image', 'picture', or is empty on an informative image	Yes
J13	E	Judgment check	Inconsistent terminology	Same concept named with different words across the unit (synonym clusters)	Yes
J14	D	Judgment check	Button label vague	'Submit', 'OK', 'Go' instead of verb + object	Yes
J15	D	Judgment check	Ambiguous references	'this', 'it', 'the above', 'below' with no clear antecedent or positional reference	Yes
N1	C	Navigation and orientation	Location cue	aria-current or highlighted nav item on every page	No
N2	C	Navigation and orientation	Breadcrumb on hierarchical pages	nav + ol + aria-current; present when depth >= 3 (depth number is an inference)	No
N3	D	Navigation and orientation	Multiple ways to find a page	>= 2 of: nav, search, sitemap, related links, table of contents	No
N4	E	Navigation and orientation	Consistent navigation order	Same nav DOM order across all pages of the unit	No
N5	E	Navigation and orientation	Consistent help position	Help / contact link in the same place across pages	No
N6	F	Navigation and orientation	Single navigation system	More than one primary nav landmark -> flag (children find redundant navs 'very confusing')	No
N7	F	Navigation and orientation	In-site Back / Home control	Visible on every page; young children do not use the browser Back button	No
N8	F	Navigation and orientation	Dropdown depth and size	<= 2 tiers; <= 6 links per dropdown; click-activated, not hover-only	No
N9	E	Navigation and orientation	Top-level item count, click depth	Report only; no sourced limit (3-click rule has no evidence)	n/a (report only)

ID	Tier	Category	Metric	Measure / threshold	Calibrate ?
N10	F	Navigation and orientation	Search has spellcheck / suggestions	Required if search exists; children misspell queries	No
N11	D	Navigation and orientation	Primary action identifiable	One visually dominant call-to-action per page	Yes
L1	C	Layout and visual	Line length	<= 80 characters; ideal 50-75	No
L2	C	Layout and visual	Line height / paragraph spacing	line-height >= 1.5; paragraph spacing >= 1.5 x line-height; no justified text	No
L3	F	Layout and visual	Body font size	>= 16 px floor; >= 18.7 px (14 pt) for younger children (inference; sources disagree)	No
L4	F	Layout and visual	Target size	>= 24 x 24 CSS px (2.2 AA); 44 x 44 (AAA); larger for children (inference). axe target-size rule is disabled by default	No
L5	D	Layout and visual	Non-text contrast	>= 3:1 for icons, borders, focus rings	No
L6	C	Layout and visual	Reflow / zoom	No 2-D scrolling at 320 CSS px; 200 % zoom without loss (needs headless browser)	No
L7	C	Layout and visual	Visual consistency of same-role elements	Computed-style variance across h2, buttons, links	No
L8	C	Layout and visual	Visual density	Interactive elements per viewport; text-to-whitespace ratio; report only	n/a (report only)
L9	D	Layout and visual	Above-the-fold signposting	Key links / previews in first viewport; no 'false floors'; attention drops ~84 % below fold	No
L10	F	Layout and visual	Scroll cue	Content visibly continues below the fold (children scroll only when cued)	No
L11	C	Layout and visual	Icons paired with text	Every icon has adjacent text (children read icons literally)	No
L12	C	Layout and visual	Number of fonts / colours	Report count; flag > 2 font families	Yes
M1	D	Motion and timing	Autoplay / motion > 5 s	Pause / stop / hide control required	No

ID	Tier	Category	Metric	Measure / threshold	Calibrate ?
M2	D	Motion and timing	Carousel	<= 5 frames; no auto-advance; children like animation, so flag rather than fail	No
M3	F	Motion and timing	Response time / feedback	Progress indication when > 1 s; children are less patient than adults	No

ID	Tier	Auto	WCAG criterion / level	Source 1	Source 1 link
T1	C	AUTO	3.1.5 (AAA) context	Kincaid et al. 1975 (Flesch-Kincaid)	https://apps.dtic.mil/sti/pdfs/ADA006655.pdf
T2	C	AUTO	3.1.5 (AAA) context	Dale-Chall formula (overview)	https://en.wikipedia.org/wiki/Dale%E2%80%93Chall_readability_formula
T3	C	AUTO	-	GOV.UK blog: 25-word sentence limit	https://insidegovuk.blog.gov.uk/2014/08/04/sentence-length-why-25-words-is-our-limit/
T4	C	AUTO	G153 (technique for 3.1.5)	WCAG technique G153	https://www.w3.org/WAI/WCAG21/Techniques/general/G153
T5	D	SEMI	WCAG 3.0 WD 'Clear language'	WCAG 3.0 Working Draft (Mar 2026)	https://www.w3.org/TR/2026/WD-wcag-3.0-20260303/
T6	C	AUTO	-	digital.gov Plain language: writing	https://digital.gov/guides/plain-language/writing
T7	C	AUTO	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/2021/NOTE-coga-usable-20210429/
T8	D	SEMI	3.1.4 (AAA); WCAG 3.0 Core	WCAG 3.1.4 Abbreviations (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/abbreviations.html
T9	D	SEMI	3.1.3 (AAA)	WCAG 3.1.3 Unusual Words (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/unusual-words.html
T10	D	SEMI	3.1.5 (AAA)	WCAG 3.1.5 Reading Level (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/reading-level.html
T11	D	External tool	-	MetaMetrics Lexile partner guide	https://www2.metametricsinc.com/hubfs/Introductory-Guide-for-Partners_Lexile-Text-Measures.pdf
S1	C	AUTO	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
S2	C	AUTO	-	Canada.ca Content Style Guide	https://design.canada.ca/style-guide/index.html

ID	Tier	Auto	WCAG criterion / level	Source 1	Source 1 link
S3	C	AUTO	2.4.10 (AAA)	WCAG 2.4.10 Section Headings (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/section-headings.html
S4	D	SEMI	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
S5	C	AUTO	-	Canada.ca Content Style Guide	https://design.canada.ca/style-guide/index.html
S6	D	SEMI	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
S7	D	SEMI	-	NN/g F-shaped pattern	https://www.nngroup.com/articles/f-shaped-pattern-reading-web-content/
J1	D	SEMI	2.4.6 (AA, 0 % automated)	WCAG 2.4.6 Headings and Labels (AA)	https://www.w3.org/WAI/WCAG22/Understanding/headings-and-labels.html
J2	D	SEMI	2.4.6 (AA)	WCAG 2.4.6 Headings and Labels (AA)	https://www.w3.org/WAI/WCAG22/Understanding/headings-and-labels.html
J3	D	SEMI	2.4.6 (AA)	WCAG 2.4.6 Headings and Labels (AA)	https://www.w3.org/WAI/WCAG22/Understanding/headings-and-labels.html
J4	D	SEMI	2.4.2 (A); 2.4.6 (AA)	WCAG 2.4.2 Page Titled (A)	https://www.w3.org/WAI/WCAG22/Understanding/page-titled.html
J5	C	AUTO	2.4.4 (A, 0 % automated)	WCAG 2.4.4 Link Purpose (In Context) (A)	https://www.w3.org/WAI/WCAG22/Understanding/link-purpose-in-context.html
J6	E	SEMI	2.4.4 (A)	WCAG 2.4.4 Link Purpose (In Context) (A)	https://www.w3.org/WAI/WCAG22/Understanding/link-purpose-in-context.html
J7	E	CRAWL	3.2.4 (AA); 2.4.4 (A)	WCAG 3.2.4 Consistent Identification (AA)	https://www.w3.org/WAI/WCAG22/Understanding/consistent-identification.html
J8	E	CRAWL	3.2.4 (AA)	WCAG 3.2.4 Consistent Identification (AA)	https://www.w3.org/WAI/WCAG22/Understanding/consistent-identification.html
J9	D	SEMI	-	NN/g Information scent	https://www.nngroup.com/articles/information-scent/

ID	Tier	Auto	WCAG criterion / level	Source 1	Source 1 link
J10	D	SEMI	1.1.1 (A) context	WCAG 1.1.1 Non-text Content	https://www.w3.org/WAI/WCAG22/Understanding/non-text-content.html
J11	C	AUTO	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
J12	D	SEMI	1.1.1 (A) quality part	WCAG 1.1.1 Non-text Content	https://www.w3.org/WAI/WCAG22/Understanding/non-text-content.html
J13	E	SEMI	3.2.4 (AA)	WCAG 3.2.4 Consistent Identification (AA)	https://www.w3.org/WAI/WCAG22/Understanding/consistent-identification.html
J14	D	SEMI	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
J15	D	SEMI	1.3.3 (A) for positional refs	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
N1	C	AUTO	2.4.8 (AAA)	WCAG 2.4.8 Location (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/location.html
N2	C	AUTO	2.4.8 (AAA)	WCAG 2.4.8 Location (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/location.html
N3	D	SEMI	2.4.5 (AA, 0 % automated)	WCAG 2.4.5 Multiple Ways (AA)	https://www.w3.org/WAI/WCAG22/Understanding/multiple-ways.html
N4	E	CRAWL	3.2.3 (AA, 0 % automated)	WCAG 3.2.3 Consistent Navigation (AA)	https://www.w3.org/WAI/WCAG22/Understanding/consistent-navigation.html
N5	E	CRAWL	3.2.6 (2.2 A)	WCAG 3.2.6 Consistent Help (2.2 A)	https://www.w3.org/WAI/WCAG22/Understanding/consistent-help.html
N6	F	AUTO	-	NN/g Children's websites usability	https://www.nngroup.com/articles/childrens-websites-usability-issues/
N7	F	AUTO	WCAG 3.0 WD 'return to start' (Developing)	NN/g Children's websites usability	https://www.nngroup.com/articles/childrens-websites-usability-issues/
N8	F	AUTO	-	NN/g Menu design	https://www.nngroup.com/articles/menu-design/
N9	E	AUTO / CRAWL	-	NN/g 3-click rule	https://www.nngroup.com/articles/3-click-rule/

ID	Tier	Auto	WCAG criterion / level	Source 1	Source 1 link
N10	F	SEMI	-	Large & Beheshti, children's web portal guidelines	https://eprints.rclis.org/8557/1/Large_web.pdf
N11	D	SEMI	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
L1	C	AUTO	1.4.8 (AAA)	WCAG 1.4.8 Visual Presentation (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/visual-presentation.html
L2	C	AUTO	1.4.8 (AAA)	WCAG 1.4.8 Visual Presentation (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/visual-presentation.html
L3	F	AUTO	-	GOV.UK Design System: Typography	https://design-system.service.gov.uk/styles/typography/
L4	F	AUTO	2.5.8 (2.2 AA); 2.5.5 (AAA)	WCAG 2.5.8 Target Size (Minimum) (2.2 AA)	https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum.html
L5	D	SEMI	1.4.11 (2.1 AA, 0 % automated)	WCAG 1.4.11 Non-text Contrast (2.1 AA)	https://www.w3.org/WAI/WCAG22/Understanding/non-text-contrast.html
L6	C	AUTO (browser)	1.4.10 (2.1 AA); 1.4.4 (AA)	WCAG 1.4.10 Reflow (2.1 AA)	https://www.w3.org/WAI/WCAG22/Understanding/reflow.html
L7	C	AUTO	COGA 4.2.3	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
L8	C	AUTO	-	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
L9	D	SEMI	-	NN/g Page fold manifesto	https://www.nngroup.com/articles/page-fold-manifesto/
L10	F	SEMI	-	NN/g Children's websites usability	https://www.nngroup.com/articles/childrens-websites-usability-issues/
L11	C	AUTO	-	Large & Beheshti, children's web portal guidelines	https://eprints.rclis.org/8557/1/Large_web.pdf
L12	C	AUTO	COGA consistency	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/
M1	D	SEMI	2.2.2 (A, often missed)	WCAG 2.2.2 Pause, Stop, Hide (A)	https://www.w3.org/WAI/WCAG22/Understanding/pause-stop-hide.html

ID	Tier	Auto	WCAG criterion / level	Source 1	Source 1 link
M2	D	SEMI	-	NN/g Designing effective carousels	https://www.nngroup.com/articles/designing-effective-carousels/
M3	F	AUTO	-	NN/g Response times	https://www.nngroup.com/articles/response-times-3-important-limits/

ID	Tier	Source 2	Source 2 link	Source 3
T1	C	Nelson et al. 2012, Measures of Text Difficulty	https://achievethecore.org/content/upload/nelson_perfetti_liben_measures_of_text_difficulty_research_ela.pdf	textstat (Python library)
T2	C	Spache formula (overview)	https://readabilityformulas.com/the-spache-readability-formula-for-young-readers/	Begeny & Greene 2014 (readability formulas, grades 2-5)
T3	C	Canada.ca Content Style Guide	https://design.canada.ca/style-guide/index.html	Ontario.ca Style Guide
T4	C			
T5	D	Nelson et al. 2012, Measures of Text Difficulty	https://achievethecore.org/content/upload/nelson_perfetti_liben_measures_of_text_difficulty_research_ela.pdf	
T6	C			
T7	C	Dale–Chall 3,000 familiar-word list (overview)	https://en.wikipedia.org/wiki/Dale%E2%80%93Chall_readability_formula	
T8	D	WCAG 3.0 Working Draft (Mar 2026)	https://www.w3.org/TR/2026/WD-wcag-3.0-20260303/	
T9	D	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/	
T10	D			
T11	D	Napolitano et al. 2015 (TextEvaluator)	https://aclanthology.org/N15-3020.pdf	Nelson et al. 2012, Measures of Text Difficulty
S1	C	Canada.ca Content Style Guide	https://design.canada.ca/style-guide/index.html	Ontario.ca Style Guide
S2	C			

ID	Tier	Source 2	Source 2 link	Source 3
S3	C			
S4	D			
S5	C			
S6	D	NN/g Children's websites usability	https://www.nngroup.com/articles/childrens-websites-usability-issues/	
S7	D			
J1	D	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/	
J2	D	NN/g F-shaped pattern	https://www.nngroup.com/articles/f-shaped-pattern-reading-web-content/	
J3	D			
J4	D	WCAG 2.4.6 Headings and Labels (AA)	https://www.w3.org/WAI/WCAG22/Understanding/headings-and-labels.html	
J5	C	NN/g Information scent	https://www.nngroup.com/articles/information-scent/	WebAIM Million 2026
J6	E	NN/g Information scent	https://www.nngroup.com/articles/information-scent/	
J7	E	WCAG 2.4.4 Link Purpose (In Context) (A)	https://www.w3.org/WAI/WCAG22/Understanding/link-purpose-in-context.html	
J8	E			
J9	D	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/	

ID	Tier	Source 2	Source 2 link	Source 3
J10	D	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/	Large & Beheshti, children's web portal guidelines
J11	C	Large & Beheshti, children's web portal guidelines	https://eprints.rclis.org/8557/1/Large_web.pdf	
J12	D	W3C ACT Rules	https://www.w3.org/WAI/standards-guidelines/act/rules/	
J13	E	W3C COGA: Making Content Usable (2021)	https://www.w3.org/TR/coga-usable/	
J14	D	GOV.UK Design System: Button	https://design-system.service.gov.uk/components/button/	
J15	D	WCAG 1.3.3 Sensory Characteristics	https://www.w3.org/WAI/WCAG22/Understanding/sensory-characteristics.html	
N1	C			
N2	C	USWDS Breadcrumb	https://designsystem.digital.gov/components/breadcrumb/	NN/g Breadcrumbs
N3	D	Deque automated coverage report	https://accessibility.deque.com/hubfs/Accessibility-Coverage-Report.pdf	
N4	E	Deque automated coverage report	https://accessibility.deque.com/hubfs/Accessibility-Coverage-Report.pdf	
N5	E			
N6	F			
N7	F	WCAG 3.0 Working Draft (Mar 2026)	https://www.w3.org/TR/2026/WD-wcag-3.0-20260303/	
N8	F	USWDS Header	https://designsystem.digital.gov/components/header/	Hourcade 2008, Interaction Design and Children
N9	E			

ID	Tier	Source 2	Source 2 link	Source 3
N10	F	NN/g Kids' cognition	https://www.nngroup.com/articles/kids-cognition/	
N11	D	NN/g Information scent	https://www.nngroup.com/articles/information-scent/	
L1	C	Baymard: Line length	https://baymard.com/blog/line-length-readability	
L2	C			
L3	F	NN/g Children's websites usability	https://www.nngroup.com/articles/childrens-websites-usability-issues/	
L4	F	WCAG 2.5.5 Target Size (Enhanced) (AAA)	https://www.w3.org/WAI/WCAG22/Understanding/target-size-enhanced.html	Hourcade 2008, Interaction Design and Children
L5	D			
L6	C	WCAG 1.4.4 Resize Text	https://www.w3.org/WAI/WCAG22/Understanding/resize-text.html	
L7	C			
L8	C	NN/g How users read on the web	https://www.nngroup.com/articles/how-users-read-on-the-web/	
L9	D			
L10	F			
L11	C			
L12	C			
M1	D			

ID	Tier	Source 2	Source 2 link	Source 3
M2	D			
M3	F	Hourcade 2008, Interaction Design and Children	https://www.cs.uic.edu/~i523/hourcade.pdf	

ID	Tier	Source 3 link
T1	C	https://pypi.org/project/textstat/
T2	C	https://news.ncsu.edu/2014/01/wms-begeny-readability
T3	C	https://www.ontario.ca/page/ontarioca-style-guide
T4	C	
T5	D	
T6	C	
T7	C	
T8	D	
T9	D	
T10	D	
T11	D	https://achievethecore.org/content/upload/nelson_perfetti_liben_measures_of_text_difficulty_research_ela.pdf
S1	C	https://www.ontario.ca/page/ontarioca-style-guide
S2	C	

ID	Tier	Source 3 link
S3	C	
S4	D	
S5	C	
S6	D	
S7	D	
J1	D	
J2	D	
J3	D	
J4	D	
J5	C	https://webaim.org/projects/million/
J6	E	
J7	E	
J8	E	
J9	D	

ID	Tier	Source 3 link
J10	D	https://eprints.rclis.org/8557/1/Large_web.pdf
J11	C	
J12	D	
J13	E	
J14	D	
J15	D	
N1	C	
N2	C	https://www.nngroup.com/articles/breadcrumbs/
N3	D	
N4	E	
N5	E	
N6	F	
N7	F	
N8	F	https://www.cs.uic.edu/~i523/hourcade.pdf
N9	E	

ID	Tier	Source 3 link
N10	F	
N11	D	
L1	C	
L2	C	
L3	F	
L4	F	https://www.cs.uic.edu/~i523/hourcade.pdf
L5	D	
L6	C	
L7	C	
L8	C	
L9	D	
L10	F	
L11	C	
L12	C	
M1	D	

ID	Tier	Source 3 link
M2	D	
M3	F	